GPT-SoVITS Guide

Train on the TUF F15 (Windows, NVIDIA) · Run on the MacBook (M2, 8 GB) · 2026-10-08

Part	Machine	Time
A. Setup and record	TUF	Day 1
B. Train	TUF	Day 2 (about 1–2 h of training)
C. Move model files	TUF → Mac	10 min
D. Install and run	Mac	Day 3
E. Test and report	Mac	Day 3

> Voice owner must agree to having their voice cloned. Before any monetized use, check the licence of the GPT-SoVITS base models on their Hugging Face page. YouTube may need the "synthetic content" label.

A. TUF: setup and record

A1. Prepare the laptop

1. Plug in the charger. Armoury Crate → mode **Turbo**. Sleep = Never (plugged in). Use a hard surface or cooling pad.
2. Keep 30 GB free. Update the NVIDIA driver (nvidia.com/drivers), restart.
3. PowerShell: `nvidia-smi`. Write down the GPU name and VRAM.
4. Install tools (PowerShell):

```
winget install --id 7zip.7zip -e
winget install --id Audacity.Audacity -e
winget install --id Gyan.FFmpeg -e
```

5. Create the work folder: `mkdir C:\voicelab\dataset`. Keep everything under `C:\voicelab` (no spaces, no non-English letters).

A2. Record the voice

1. Quiet room, microphone 10–15 cm from the mouth, no music, no effects, no noise-reduction in the recording app.
2. Audacity: project rate **44100 Hz**, **Mono**. Peaks between -12 dB and -6 dB, never clipping.
3. Record **20–30 minutes** total (minimum 10):
    - 10 min: the sentences below.
    - 10 min: a public-domain book (Project Gutenberg).
    - 10 min: a draft video script in the channel style.

```
1.  Welcome back. Today we are going to build something that runs while you sleep.
2.  Wait, really? That actually worked on the first try?
3.  Okay, let me explain this slowly, because it is easy to miss.
4.  The workflow has three steps: collect the data, clean it, and send the report.
5.  I did not expect that. Honestly, I am a little bit surprised.
6.  Never trust a number you have not checked twice.
7.  Here is the problem: the server answers, but it answers with nothing.
8.  Great. Now we press the button, and we wait.
9.  It costs zero dollars, and it takes about ten minutes.
10. Don't worry, we will fix it together. It is just a missing comma.
11. Is that a bug, or is it a feature? Let's find out.
12. In March, two hundred and forty people signed up; by June, there were over
three thousand.
13. Blender, n8n, Docker, and a language model walk into a car.
14. Quiet, now. Something is happening on the screen.
15. That is the whole trick. Nothing more, nothing less.
16. Okay, okay, that was a bad joke.
17. If it fails again, do not panic. Read the error message out loud.
18. The camera moves left, the road bends, and the sun goes down behind the hills.
19. Why would anyone build it this way? Because it works, and because it is ours.
20. Thank you for watching. See you in the next one.
```

4. One sentence per take, 1 second of silence before and after. If you stumble, wait 2 seconds and repeat the sentence.

A3. Clean and export

1. Delete mistakes, coughs, clicks and silences longer than 1.5 s.
2. Effect → Normalize to **-3 dB**. No compression, no EQ. Noise reduction only if the room was noisy (mild, 6–12 dB).
3. Export WAV, 44100 Hz, mono, 16-bit into `C:\voicelab\dataset` as `take_001.wav`, `take_002.wav`, … Back the folder up.
4. Make two **reference clips**, each **5–9 seconds** (the tool rejects anything outside 3–10 s), clean, no stumbles:

```
C:\voicelab\ref_neutral.wav   + C:\voicelab\ref_neutral.txt   (exact words)
C:\voicelab\ref_excited.wav   + C:\voicelab\ref_excited.txt   (exact words)
```

5. Listen to 5 random files with headphones: no hiss, echo, clipping or other voices. Re-record if many fail.

B. TUF: install and train

B1. Install

1. Open github.com/RVC-Boss/GPT-SoVITS. In the README, open the Windows integrated package link (Hugging Face repo `lj1995/GPT-SoVITS-windows-package`). Download the newest `.7z` (several GB).
2. Right-click → 7-Zip → Extract to `C:\voicelab\GPT-SoVITS`.
3. Double-click `go-webui.bat`. Keep the black console open. The browser opens the WebUI (usually http://localhost:9874).
4. If Windows Defender blocks files, confirm the source is the official repo, then add `C:\voicelab` as an exclusion.

B2. Prepare the dataset

WebUI tab **1-GPT-SoVITS-TTS**. In the version dropdown choose the newest (v2 or later).

1. **0b. Audio slicer:** input `C:\voicelab\dataset`, output `C:\voicelab\slices`, defaults. Click **Start slicing**. Wait for "done".
2. **0c. ASR:** input folder `C:\voicelab\slices`, model **Faster Whisper**, language **en**, precision float16. Click **Start**. It writes a `.list` file. Note its path from the log.
3. **0d. Proofreading:** tick "Open labelling WebUI", fix every wrong word, delete lines with noise, click **Submit Text**, then **Save File**. Wrong text = wrong pronunciation. Take 15–30 min.
4. **1A. Dataset formatting:** experiment name `myvoice`; text file = the `.list` path; audio folder = `C:\voicelab\slices`. Click **Start one-click formatting**. Wait until the three steps (tokenization, SSL feature extraction, semantic tokens) all finish without red errors.

B3. Train

Tab **1B-Fine-tuning**. Pick batch sizes by VRAM:

VRAM	SoVITS batch size	GPT batch size
4 GB	2	2
6 GB	4	4
8 GB	8	6

1. **SoVITS training:** batch size from the table, total epochs **8**, save every **4**, "save only latest" **Off**, "save small model to weights" **On**. Click **Start SoVITS training**. 10–40 min.
2. **GPT training:** batch size from the table, total epochs **15**, save every **5**, "save only latest" **Off**, "save weights" **On**, DPO off. Click **Start GPT training**. 10–40 min.
3. Watch VRAM with `nvidia-smi -l 2`. On "CUDA out of memory": lower the batch size by half, restart that step.
4. Finished weights are in the package folder:

```
C:\voicelab\GPT-SoVITS\GPT_weights_v2\myvoice-e15.ckpt   (and other epochs)
C:\voicelab\GPT-SoVITS\SoVITS_weights_v2\myvoice_e8_s....pth
```

(folder names carry the version, e.g. `_v2`, `_v2Pro`.)

B4. Quick check on the TUF

1. Tab **1C-inference**: tick **Open TTS inference WebUI**, click refresh, choose the highest-epoch GPT and SoVITS weights.
2. Upload `ref_neutral.wav`, paste the exact text, language English. Generate T01 and T02 from section E. If it sounds clearly wrong, retrain with +4 SoVITS epochs or re-check the transcripts in 0d.

C. Move the model to the Mac

1. Copy to a USB drive or cloud folder, keeping names:

```
GPT_weights_v2\myvoice-e15.ckpt (the best 1–2 epochs)
SoVITS_weights_v2\myvoice_e8_s....pth (the best 1–2 epochs)
ref_neutral.wav + ref_neutral.txt
ref_excited.wav + ref_excited.txt
```

2. On the Mac, put them in `~/voicelab/model/`.

D. Mac: install and run

Open Terminal. If the README (github.com/RVC-Boss/GPT-SoVITS) gives different install lines, follow the README.

D1. Prerequisites

```
xcode-select --install
brew install ffmpeg
brew install --cask miniconda
conda init zsh
```

Close and reopen Terminal. Free at least 15 GB disk. Close Chrome and other heavy apps while running.

D2. Install

```
cd ~/voicelab
git clone https://github.com/RVC-Boss/GPT-SoVITS.git
cd GPT-SoVITS
conda create -n GPTSoVits python=3.10 -y
conda activate GPTSoVits
bash install.sh --device MPS --source HF
```

This downloads the pretrained base models (several GB). If it ends with an error about MPS, rerun with `--device CPU`.

D3. Place your model

```
mkdir -p GPT_weights_v2 SoVITS_weights_v2
cp ~/voicelab/model/myvoice-e15.ckpt GPT_weights_v2/
cp ~/voicelab/model/myvoice_e8_s*.pth SoVITS_weights_v2/
```

Use the same version suffix (`_v2`, `_v2Pro`, …) that the TUF used.

D4. Run with the web UI

```
cd ~/voicelab/GPT-SoVITS
conda activate GPTSoVits
python webui.py
```

1. Open the printed address (usually http://localhost:9874). Tab **1C-inference**, tick **Open TTS inference WebUI**.
2. Refresh, choose your GPT and SoVITS weights.
3. Upload `ref_neutral.wav`, paste its exact text, language English. Paste the text to speak. Click **Synthesize**.
4. If you get a half-precision or MPS error, tick "No half precision" (or switch to CPU, see D2).

D5. Run as an API (for n8n and scripts)

```
cd ~/voicelab/GPT-SoVITS
conda activate GPTSoVits
python api_v2.py -a 127.0.0.1 -p 9880 -c GPT_SoVITS/configs/tts_infer.yaml
```

Switch to your model once after the server starts:

```
curl "http://127.0.0.1:9880/set_gpt_weights?weights_path=GPT_weights_v2/myvoice-e15.ckpt"
curl "http://127.0.0.1:9880/set_sovits_weights?
weights_path=SoVITS_weights_v2/myvoice_e8_s....pth"
```

Generate a line (replace the weights file and text):

```
curl -X POST http://127.0.0.1:9880/tts -H "Content-Type: application/json" -d '{
 "text":"Welcome back. Today we are going to build something that runs while you sleep.",
 "text_lang":"en",
 "ref_audio_path":"/Users/YOU/voicelab/model/ref_neutral.wav",
 "prompt_text":"EXACT TEXT OF ref_neutral",
 "prompt_lang":"en",
 "media_type":"wav"}' --output T01.wav
```

In n8n use an HTTP Request node with the same JSON body and the response set to "File".

E. Test and report

E1. Test lines (generate all 10)

```
T01. Welcome back. Today we are going to build something that runs while you sleep.
T02. Wait, really? That actually worked on the first try?
T03. Okay, let me explain this slowly, because it is easy to miss.
T04. In March, two hundred and forty people signed up; by June, there were over three
thousand.
T05. Blender, n8n, Docker, and a language model walk into a car.
T06. Don't worry, we will fix it together. It is just a missing comma.
T07. The camera moves left, the road bends, and the sun goes down behind the hills.
T08. Honestly? I did not expect that. I am a little bit surprised, and a little bit
scared.
T09. Here is the problem: the server answers, but it answers with nothing. So what do we
do? We check the logs, one line at a time, until the answer shows up.
T10. Thank you for watching. See you in the next one.
```

Generate each with: (a) `ref_neutral`, (b) `ref_excited`, (c) two different epochs of the model. Name files `T01_neutral_e15.wav` etc.

E2. Tuning

Problem	Fix
Words repeated or skipped	top_k 5–10, temperature 0.6–0.8, split the sentence
Flat, robotic delivery	Try `ref_excited`; use a reference with the emotion you want
Wrong pronunciation of a word	Spell it phonetically in the text; write numbers as words
Metallic or glitchy	Use an earlier epoch; check dataset noise and transcripts
Speech cut at the end	Add a period; reduce the text length per call

E3. Measure on the Mac

- Seconds to generate T09 (length of audio vs time taken).
- Memory: Activity Monitor → Memory tab, peak while generating. Does it swap?
- Same voice on the 10 lines? Any glitches?

F. Troubleshooting

Symptom	Fix
TUF: CUDA out of memory	Halve batch size, close other apps, restart step
TUF: slicing or ASR fails	Check the folder path has no spaces; check `ffmpeg -version` in a new PowerShell
TUF: console closes immediately	Run `go-webui.bat` from PowerShell to read the error; extract again to a short path
TUF: laptop throttles	Cooling pad, Turbo mode, keep checkpoints saved
Mac: install.sh fails	Check `conda activate GPTSoVits`, Python 3.10, Xcode tools installed; retry with `--device CPU`
Mac: very slow or swapping	Close all apps, use CPU, generate shorter lines
Mac: model not listed	Files must be in `GPT_weights_v2` / `SoVITS_weights_v2` (same version suffix); click refresh
Mac: reference rejected	Reference clip must be 3–10 seconds

G. Send back

□ Trained GPT and SoVITS weight files and both reference clips with their texts.
□ All generated test lines.
□ Dataset notes: minutes recorded, microphone, room.
□ Measurements from E3 and the TUF training times and VRAM use.
□ The exact commands that worked, and anything that failed.